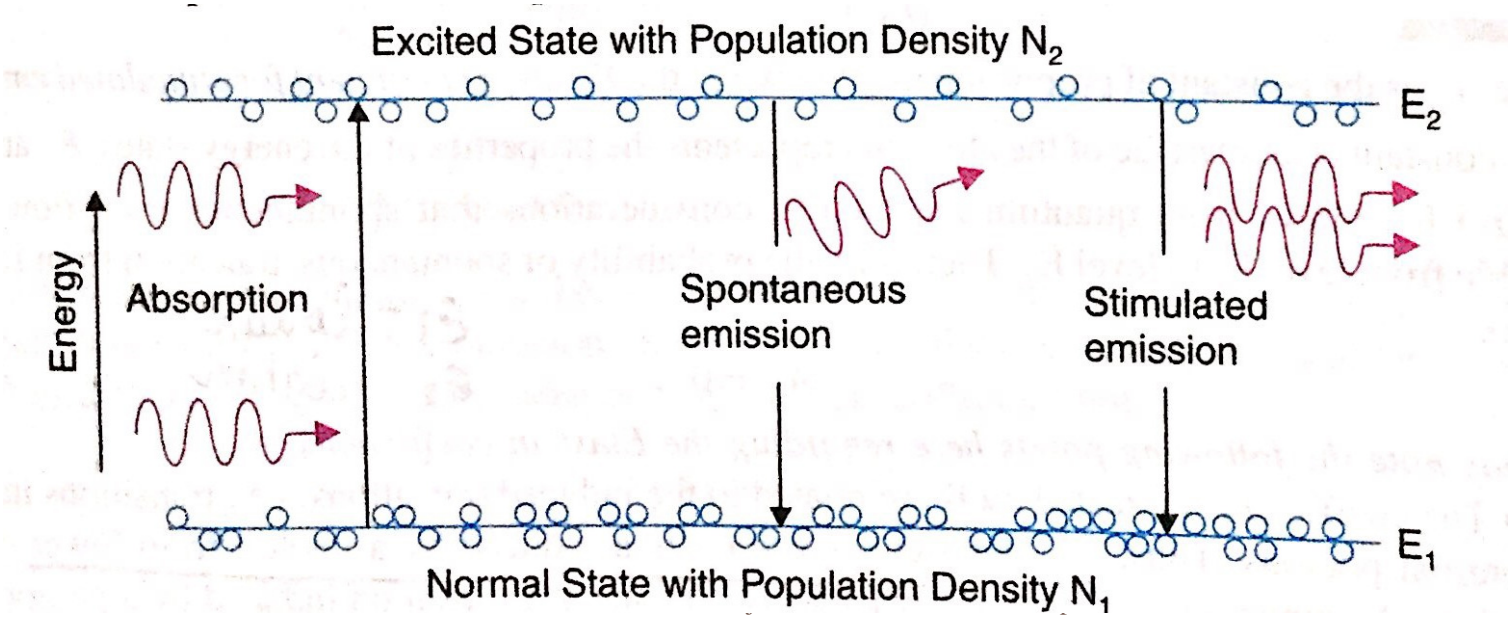
UNIT 2: LASER AND APPLICATIONS

LECTURE 5

Revision Lecture 1

When a photon of energy (hν) travel through a material three different processes occurs

1. **Induced Absorption,** $R_{abs} = P_{12}N_1 = B_{12}\,\rho(\nu)N_1$
2. **Spontaneous Emission** $R_{sp} = P_{21}N_2 = A_{21}N_2$
3. **Stimulated Emission** $R_{st} = (P_{21})_{st}N_2 = B_{21}\,\rho(\nu)N_2$



Revision Lecture 2

Populations of energy levels is given by Boltzmann's distribution

$$\frac{N_1}{N_2} = \frac{e^{-E_1/kT}}{e^{-E_2/kT}} = e^{(E_2-E_1)/kT}$$

$$R_{ab} = R_{sp} + R_{st}$$

Absorption rate, R_{ab}
Stimulated Emission rate, R_{st}
Spontaneous Emission rate, R_{sp}

$$B_{12}\,\rho(\nu)N_1 = A_{21}\,N_2 + B_{21}\,\rho(\nu)N_2$$

$$\rho(\nu) = \frac{A_{21}\,N_2}{[B_{12}\,N_1 - B_{21}N_2]}$$

$$\rho(\nu) = \frac{A_{21}}{B_{12}}\frac{1}{[e^{h\nu/kT} - B_{21}/B_{12}]}$$

Photon density (ρ)

$$\rho(\nu) = \left[\frac{8\pi h\nu^3\mu^3}{c^3}\right]\left[\frac{1}{[e^{h\nu/kT} - 1]}\right]$$

$$\frac{A_{21}}{B_{12}} = \frac{8\pi h\nu^3\mu^3}{c^3}$$

Planck's law

Einstein relations

$$\frac{B_{21}}{B_{12}} = 1$$

Revision Lecture 2

$$R_1 = \frac{R_{st}}{R_{sp}} : \quad R_2 = \frac{R_{st}}{R_{abs}}$$

Ratios of stimulated emission rate to rate of spontaneous emission and rate of induced absorption will give us an Idea about what we need to realize **LASER**

To have LASER action dominant Stimulated emission is required

For that, we have to have high R_1 and R_2

$$R_2 = \frac{N_2}{N_1}$$ ----------→ 1. Population inversion, $N_2 >> N_1$

2. High photon density (ρ)
3. Metastable state $B_{21} > A_{21}$ ←---------- $R_1 = \frac{B_{21}}{A_{21}}\rho(\nu)$

1. POPULATION INVERSION by Pumping

2. METASTABLE STATE to increase life time in the excited state

3. Large photon density (ρ) –Optical resonant cavity and photon confinement

Revision Lecture 3

Components of LASER

1. Active medium- *metastable state to reduce* A_{21}
2. Pump- *population inversion to realize* $N_2 >> N_1$
3. Optical resonant cavity- *confinement to increase* ρ

$$\gamma \geq \alpha_s + \frac{1}{2L} \ln \frac{1}{r_1 r_2}$$

Known as the condition for Lasing

$$\gamma_{th} = \alpha_s + \frac{1}{2L} \ln \frac{1}{r_1 r_2} \quad \text{Eq.2}$$

Known as the threshold condition for commencing Lasing

Revision Lecture 4

Important characteristics of the laser beam

- **Coherence**
- **Monochromaticity**
- **Collimation**
  - **Directionality**
  - **Negligible divergence**
  - **High intensity**

First Laser: Ruby laser.; a solid state laser with three level system

Types of Lasers

1. Solid state laser
   a) Ruby laser (first laser)
   b) Nd-YAG Laser
2. Gas Laser
   a) CO_2 laser
   b) CO Laser
   c) Nitrogen laser
   d) Hydrogen laser
   e)He-Ne Laser (first gas laser)
3. Liquid laser
   Dye laser
4. Semiconductor laser
   Laser diode- pn junction

Quick Quiz

Quick Quiz Response on the 9/05/2018 Lecture

No	Question	Attempts	Right	Wrong
1	Important characteristics of the laser beam's are	8	3	5
2	In the following which is a three level laser system	8	2	6
3	In the following which is a four level laser system	8	6	2

Important characteristics of the laser beam's are

a) **Coherence**

b) **Monochromaticity**

c) **Collimation**

d) **All of the above**

e) **None of the above**

Ans: D

In the following which is a three level laser system

a) Ruby Laser
b) CO_2 Laser
c) He-Ne Laser
d) Nd-YAG Laser

Ans: A

In the following which is a four level laser system

a) CO_2 Laser
b) He-Ne Laser
c) Nd-YAG Laser
d) All of the above
e) None of the above

Ans: D

Re-vision

Fundamentals of laser

$$E_2 - E_1 = h\nu$$

$$A + h\nu = A^*$$

$$A^* \rightarrow A + h\nu$$

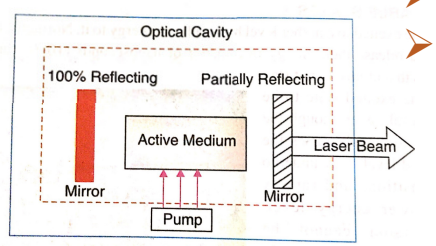
$$A^* + h\nu \rightarrow A + 2h\nu$$

$$\rho(\nu) = \left[\frac{8\pi h\nu^3\mu^3}{c^3}\right]\left[\frac{1}{[e^{h\nu/kT} - 1]}\right]$$

$$B_{12} = B_{21} = \frac{c^3}{8\pi h\nu^3\mu^3} A_{21}$$

$$\rho(\nu) = \frac{A_{21}}{B_{12}} \frac{1}{[e^{h\nu/kT} - B_{21}/B_{12}]}$$

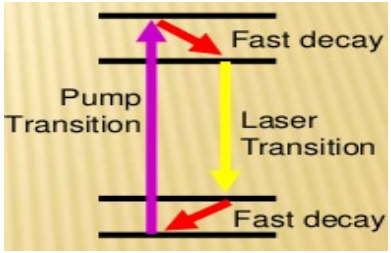
$$R_1 = \frac{B_{21}}{A_{21}}\rho(\nu) \qquad R_2 = \frac{N_2}{N_1}$$



1. Active medium- *metastable state to reduce A_{21}*
2. Pump- *population inversion to realize $N_2 >> N_1$*
3. Optical resonant cavity- *confinement to increase* $\boldsymbol{\rho}$

$$\gamma \geq \alpha_s + \frac{1}{2L}\ln\frac{1}{r_1 r_2}$$

- **Coherent**
- **Monochromatic**
- **Collimated**



(i) Large photon density (ρ) –Optical resonant cavity
(ii) Large life time of atoms in the excited state- Metastable state
(iii) Large number of excited atoms - population inversion

$$\frac{N_2}{N_1} = e^{-(E_2 - E_1)/kT}$$

Metastable state- Active centers, Lasing medium

Optical resonator, oscillator, positive feed back, mirrors

Types of Lasers

1. Solid state laser
   Nd-YAG Laser
2. Gas Laser
   He-Ne Laser (first gas laser)
3. Liquid laser
   Dye laser
4. Semiconductor laser
   Laser diode- pn junction **electrons-holes recombination**

LECTURE SCHEDULE

29/8/18 Lecture 1:*Fundamentals of laser- energy levels in atoms,* Radiation matter interaction, Absorption of light, Spontaneous emission of light, stimulated emission of light, *Einstein A and B coefficients*

30/8/18 Lecture 2: *Fundamentals of laser-* Einstein's relation, Population of energy levels, Metastable state, population inversion, Resonant cavity.

31/8/18 Lecture 3: Fundamentals of laser- Components of Laser, Excitation mechanisms, lasing action, Properties of laser,

5/9/18 Lecture 4: Type of Lasers: Nd - YAG, He-Ne Laser, Semiconductor Laser,

6/9/18 Lecture 5: Applications of laser in engineering, Holography.

Applications of laser

LASERs are used everywhere

1. **Science**

   - To study the properties of material, structure, Raman scattering, material synthesis, thin film preparation by laser ablation, heat treatment for hardening metals, annealing/sintering for densification/phase formation of material.. To study the atmospheric pollutants distribution and identification

2. **Industry**

- Used for welding, cutting of materials, machining and drilling holes
- Electronic industry: information processing, communication (fiber optic), Information storage (holography), optical memory cards
- In printers, barcode scanners, CD players,
- Guiding missiles to the target

3. **Medicine**

   - Eye surgery, treatment of dental decay, skin diseases

Holography

WHAT IS HOLOGRAPHY?

- Holography (from the Greek, whole + write) is the science of producing holograms
- It is an advanced form of photography that allows an image to be recorded in three dimensions.
- The technique of holography can also be used to optically store, retrieve, and process information.

Holography

HISTORY

- Dennis Gabor - Father of Holography and Holographic Technologies
- Dennis wrote a paper in 1948 that has become the foundation of modern Holography.
- The most interesting thing about all this is that laser light had not even been invented yet, when he wrote his paper.

Gabor: Nobel Prize 1971

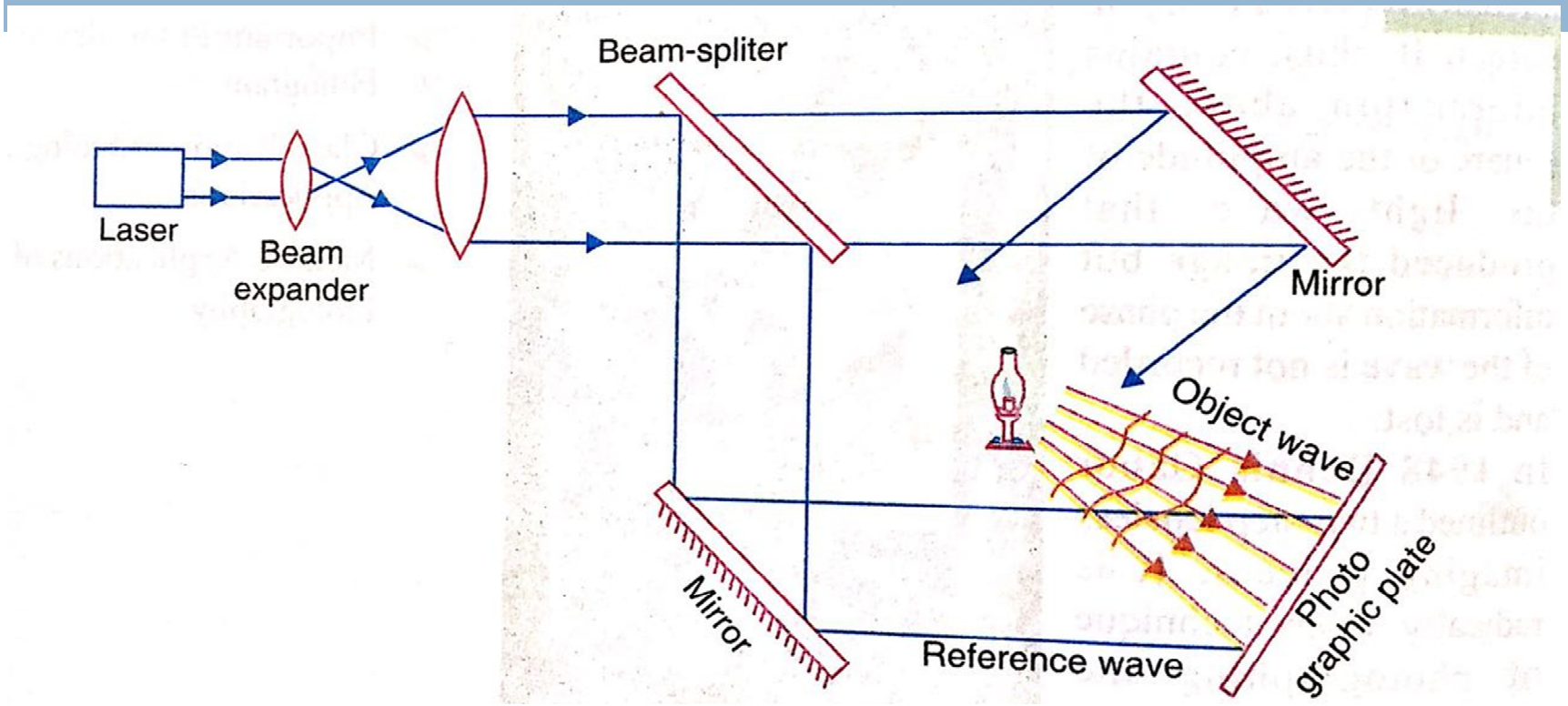
Holography

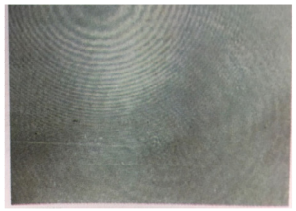
Is a two-step process

1. Recording of the hologram
2. Reconstruction of the image from the hologram

- No lenses are required compared to conventional photography. But need other more complicated procedures☺.
- Coherent wave required--- so Laser is needed
- A reference beam also required along with the laser beam scattered from the object to be imaged– so a beam splitter needed
- Two mirrors to reflect the beam to photographic plate
- The photographic plate is a photosensitive material like in conventional photography

Holography-recording



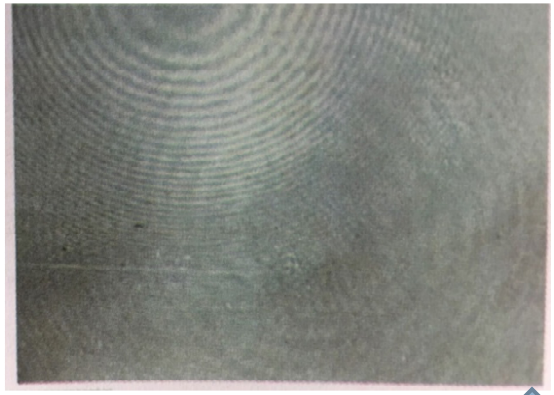


Object wave is spherical wave front
Reference wave is plane wave front

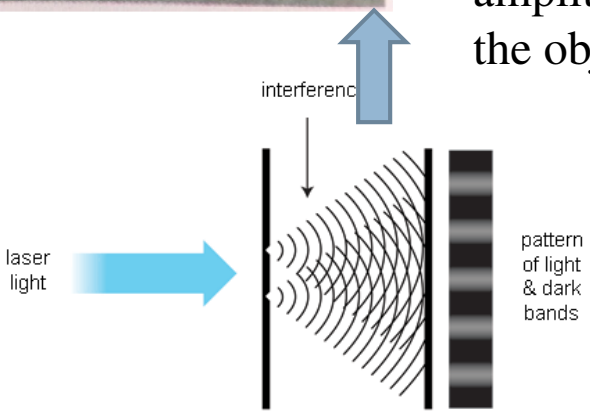
Both interfere on the photographic plane surface

Holography- Recording

Hologram



Developed negative is the Hologram.. formed accordance with the light intensity (**amplitude**) reflected from the object and **phase**.. Three dimensional information.. No distinct image of the object.
Just the information about the amplitude and phase at each point of the object



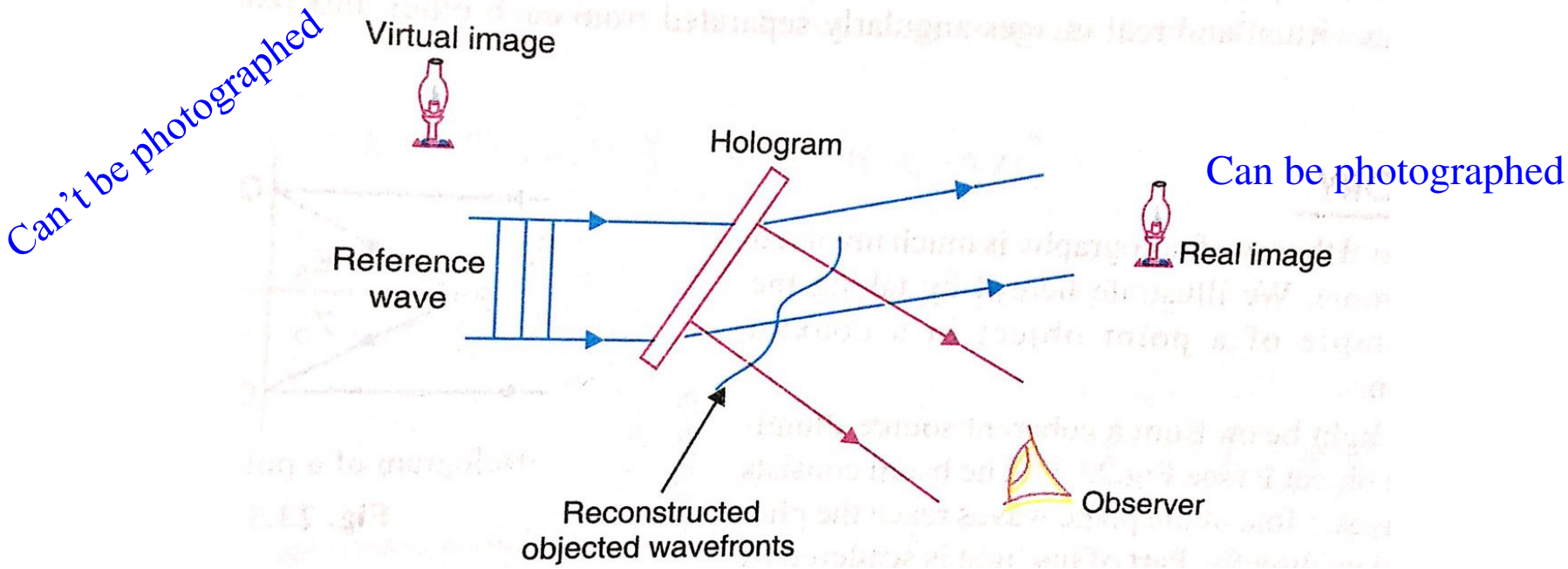
Negative image is formed accordance with the light intensity (**amplitude only**) reflected from the object.. Two dimensional information

Interference pattern

A small portion contain all information of the object?

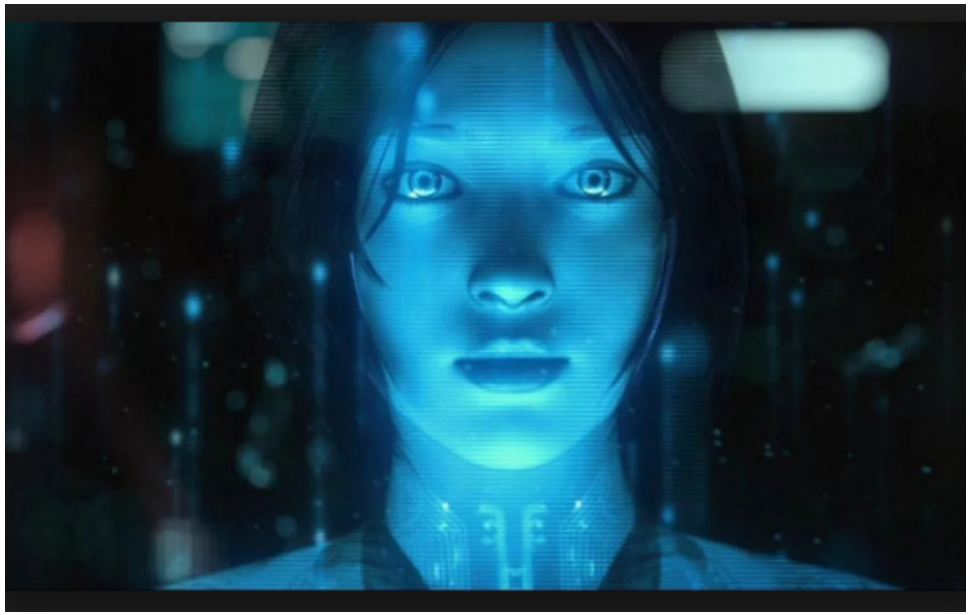
Holography-Reconstruction

For viewing the image – need reconstruction with the recorded hologram



Interference fringes acts as diffraction grating

Holography-Reconstruction



Hologram image photographed

Image developed from the negative

Holography-Applications

Medical applications

- Ophthalmology
- Endoscopy
- Otology
- Orthopedics

Holographic interferometry is used for **non-destructive testing of materials**

Information storage, optical data storage

Quick QUIZ

. Which of the following is a unique property of laser?

a) Directional
b) Speed
c) Coherence
d) Wavelength

Which of the following is an example of optical pumping?

a) Ruby laser
b) Helium-Neon laser
c) Semiconductor laser
d) Dye laser

The image produced by holography is

(a)One dimensional
(b)Two dimensional
(c)Three dimensional
(d)None of these

Hologram contains the information about

(a)Amplitude of the object
(b)Phase of the object
(c)Both amplitude and phase of the object

Hologram is the result of

(a)interference of object and reference beam
(b)polarization of the object and reference beam
(c)diffraction of the object and reference beam
(d)both (a) and (b)

Syllabus

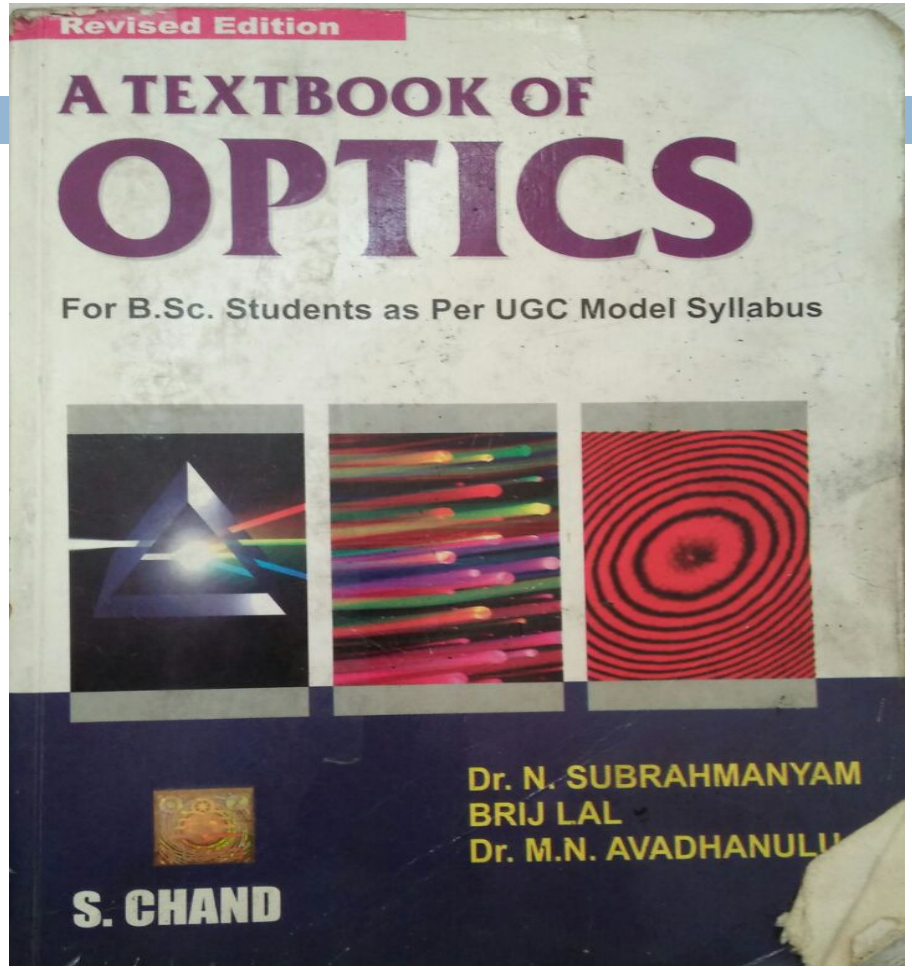
UNIT II: Lasers and applications

Fundamentals of laser- *Energy levels in atoms, Radiation matter interaction, Absorption of light, spontaneous emission of light, stimulated emission of light, Einstein A and B coefficients, Population of energy levels, Einstein's relations, metastable state, population inversion, resonant cavity, excitation mechanisms, lasing action, properties of laser*

Types of Lasers: *Nd - YAG, He-Ne Laser, Semiconductor Laser,*

Applications of laser in engineering: *Holography*.

Don't know whether I '*uncovered*' it so that so you got the real feeling of LASER.. All the best.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